

Enhancing Memory Bandwidth and Energy Efficiency in GPU Microarchitecture through High Bandwidth Memory (HBM) and Data Locality Optimizations

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Abstract—The exponential rise in data-intensive applications, such as AI, high-performance computing, and real-time analytics, has propelled the evolution of GPU microarchitecture. This paper investigates key advancements aimed at overcoming bottlenecks in memory bandwidth, energy efficiency, and control flow divergence. High Bandwidth Memory (HBM) emerges as a cornerstone technology, with innovations like 3D stacking and hybrid memory systems addressing capacity and scalability challenges. Techniques such as Thread Block Compaction (TBC), Dynamic Warp Formation (DWF), and locality-aware hierarchies enhance SIMT efficiency and optimize resource utilization. Furthermore, chiplet-based designs and logic-layer integration within HBM advance energy efficiency by reducing data traversal latency and improving thermal stability. By synthesizing these innovations, this work not only highlights the current state of GPU advancements but also outlines future research directions, including AI-driven scheduling, hybrid memory integration, and real-time thermal management, which are critical for next-generation GPUs to meet the growing demands of modern computing workloads.

Index Terms—GPU microarchitecture, High Bandwidth Memory (HBM), energy efficiency, 3D stacking, Single Instruction Multiple Thread (SIMT), Thread Block Compaction (TBC), Dynamic Warp Formation (DWF), data locality optimization, chiplet-based architectures, hybrid memory systems, AI-driven scheduling, thermal management

I. INTRODUCTION

The exponential growth in data-intensive applications, such as artificial intelligence (AI), high-performance computing (HPC), and real-time analytics, has driven unprecedented demands on modern GPU architectures. GPUs have become indispensable for accelerating computations across diverse domains, from training deep neural networks to executing complex scientific simulations. However, as workloads scale, so do the challenges of meeting performance, energy efficiency, and memory bandwidth requirements. At the heart of these challenges lies a complex interplay of factors. GPUs must efficiently manage control flow divergence in SIMT (Single Instruction, Multiple Threads) architectures, address memory bandwidth bottlenecks with innovations like High-Bandwidth Memory (HBM), and tackle thermal issues in advanced stacking technologies. If unresolved, these constraints limit the scalability and versatility of GPUs in environments ranging from dense data centres to edge computing devices. This paper aims to address these pressing challenges by synthesizing recent advancements in GPU microarchitecture. Specifically, it explores innovations in three key areas: memory hierarchy

optimization, SIMT efficiency, and energy-efficient design scalability. We focus on the integration of HBM technologies, advanced 2.5D and 3D stacking architectures, and the emerging synergy between chiplet-based designs and tailored workload management techniques. By connecting these innovations to real-world applications, such as AI inference and graph analytics, this study highlights the current progress and the potential pathways for future GPU development. Additionally, this paper bridges these innovations to address real-world GPU challenges, aiming to provide actionable insights for both current technologies and future GPU architectures. Unlike prior work, this paper synthesizes advancements in HBM integration, SIMT optimization, and chiplet synergy, providing a holistic view of GPU architectural improvements

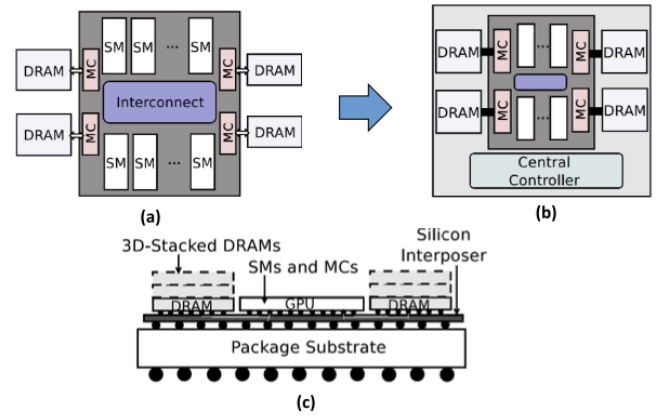


Fig. 1. Architecture of HBM-enabled GPU. (a) Conventional GPU with off-chip GDDR5 DRAM. (b) HBM-enabled GPU (top view). (c) HBM-enabled GPU (side view). [1]

This figure illustrates the transition from conventional GPU architectures using off-chip GDDR DRAM to HBM-enabled GPUs, highlighting top and side views. Subfigure (a) shows a traditional GPU design where memory controllers (MCs) interface with external DRAM modules, leading to higher latency and energy overheads due to off-chip data transfers. Subfigure (b) introduces a centralized controller in HBM-enabled GPUs, optimizing data flow and resource management. Subfigure (c) further emphasizes the 2.5D integration using a silicon interposer and vertically stacked DRAMs, significantly enhancing memory bandwidth and reducing latency. This architectural

shift underpins the focus of this paper on leveraging HBM for improved performance and energy efficiency.

II. EVOLUTION OF GPU MEMORY ARCHITECTURES

The development of modern GPU memory architectures is deeply rooted in the foundational Von Neumann architecture, introduced in the mid-20th century. This design, characterized by its central processing unit (CPU) and separate memory units communicating over a shared bus, laid the groundwork for the computing systems we see today. While highly influential, the Von Neumann architecture suffered from a significant bottleneck: the limited bandwidth between the CPU and memory, which constrained data-intensive applications.[2]

The Shift to High-Performance Memory Systems: As computational workloads became more demanding, particularly with the advent of graphics processing units (GPUs) in the 1990s, the need for more efficient memory architectures grew. Early GPUs relied on Double Data Rate Synchronous DRAM (DDR SDRAM), which provided incremental improvements in bandwidth. However, the growing complexity of applications, including 3D rendering and gaming, quickly outpaced the capabilities of DDR memory.

Graphics Double Data Rate (GDDR) memory emerged as a specialized solution, offering higher bandwidth and lower latency than traditional DDR memory. GDDR5, for instance, achieved bandwidths up to 28 GB/s per module, making it a popular choice for GPUs in the 2010s [3]. Despite these advancements, the increasing demands of high-performance computing (HPC) and AI workloads exposed GDDR's limitations, particularly in energy efficiency and scalability.

In response, the introduction of High Bandwidth Memory (HBM) marked a significant advancement. By employing Through-Silicon Vias (TSVs) and stacking DRAM dies vertically, HBM achieved unprecedented bandwidth and energy efficiency. For instance, HBM2 offered up to 256 GB/s per stack, while the more recent HBM3E delivers bandwidths exceeding 1 TB/s [4]. This architectural innovation not only alleviated previous bandwidth limitations but also introduced new challenges in thermal management and scalability.

This historical journey underscores the relentless pursuit of overcoming memory bottlenecks, transitioning from the foundational Von Neumann architecture to cutting-edge innovations like HBM and advanced stacking technologies. Each stage of development reflects the growing need for more efficient and scalable solutions, driven by the increasing complexity of GPU workloads in AI, HPC, and data-intensive applications.

As GPUs evolve, critical challenges persist: optimizing memory hierarchies to balance bandwidth and latency, improving thread utilization in SIMT architectures, and addressing thermal constraints in densely packed designs. These challenges demand innovative approaches that integrate advancements in memory architecture, computational efficiency, and energy scalability.

Building on this foundation, the subsequent sections of this paper delve into the critical concepts shaping modern

GPU microarchitecture. By exploring innovations in memory hierarchy optimization, SIMT efficiency, and energy-efficient design scalability, this study aims to provide actionable insights into how GPUs can continue to meet the demands of next-generation computing.

III. KEY INNOVATIONS AND ANALYSIS

This section explores the advancements in GPU microarchitecture across three critical domains, providing an in-depth analysis of how these innovations address contemporary challenges.

A. Memory Hierarchy Optimization

1) Role of High Bandwidth Memory (HBM): The evolution of GPU memory architectures has culminated in the development of HBM, a technology that leverages vertical stacking and Through-Silicon Vias (TSVs) to deliver unprecedented bandwidth and energy efficiency. By bringing memory closer to the processor, HBM addresses the latency and bandwidth bottlenecks inherent in earlier architectures like GDDR. For instance, HBM2 provides a peak bandwidth of 256 GB/s per stack, while the latest HBM3E surpasses 1 TB/s [5].

Performance Example: HBM-enabled GPUs demonstrated remarkable gains in AlexNet training, reducing memory consumption from 4.35 GB to 3.52 GB and achieving a 1.63 $\times$ speedup through double-buffering pipelines [1].

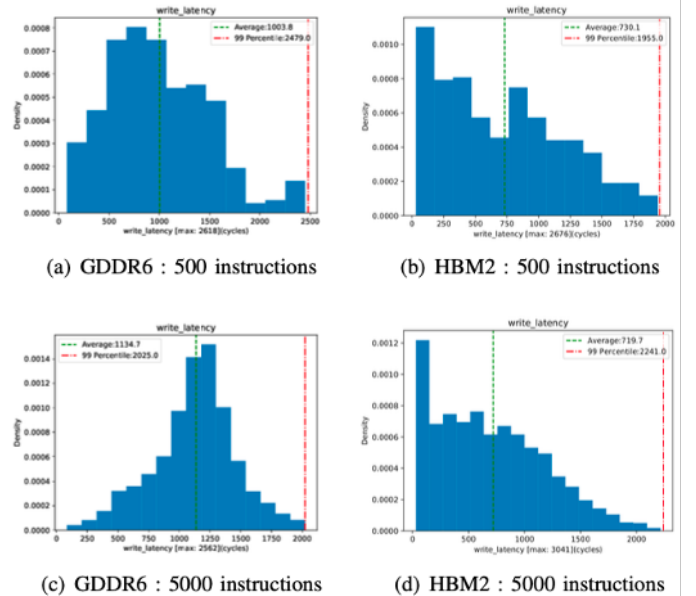


Fig. 2. Evaluation of GDDR6 and HBM across different number of write instructions [5]

Latency comparison between GDDR6 and HBM2 for varying numbers of write instructions. HBM2 demonstrates significantly lower and more stable latency, underscoring its suitability for scaling GPU architectures.

2) *2.5D and 3D Stacking Technologies*: HBM's success is underpinned by advancements in stacking technologies, which enable higher memory density and bandwidth while addressing latency and energy constraints.

(a) **2.5D Stacking**: This approach uses a silicon interposer to connect memory and logic components, offering a balance between performance and thermal management. For instance, NVIDIA's A100 GPU leverages 2.5D stacking to achieve high performance in AI and HPC applications. [6]

(b) **3D Stacking**: Vertically stacking DRAM dies provides even greater bandwidth and density. However, the increased power density poses thermal challenges, requiring innovations such as Molded Underfill (MR-MUF) and liquid cooling systems for effective heat dissipation [7]

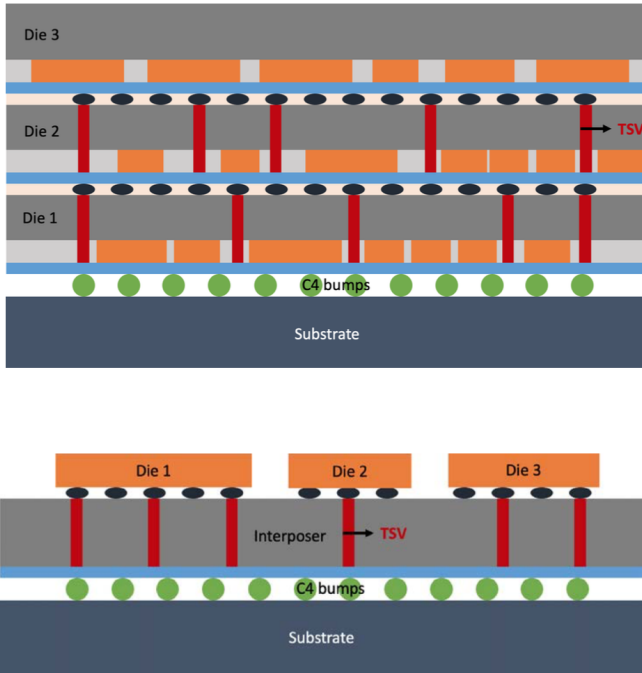


Fig. 3. Vertical cross-sections of a (a) stacked 3-D IC and (b) interposer-based 2.5-D IC.[6]

3) *Hybrid Memory Systems*: Hybrid memory architectures integrate HBM with emerging non-volatile memory (NVM) technologies, such as ReRAM or Storage-Class Memory (SCM), to address scalability and cost concerns.[8]

Advantages:

- **Scalability**: NVM technologies complement HBM by offering higher capacity at lower cost.
- **Energy Efficiency**: By offloading non-critical data to NVM, hybrid systems reduce idle power consumption by up to 30%. [8]

Example: Hybrid memory stacks dynamically allocate high-throughput workloads to HBM while relegating less time-sensitive tasks to SCM, optimizing energy and performance simultaneously [7].

4) *Locality-Aware Hierarchies*:: The Locality-Aware Memory Architecture (LAMAR) improves GPU memory efficiency

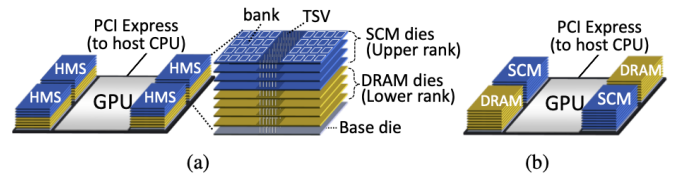


Fig. 4. Design space of a GPU with SCM and DRAM cache with (a) 3D-stacked DRAM and SCM and (b) separate DRAM and SCM stacks.[7]

by dynamically adjusting memory access granularity. Using Granularity Decision Units (GDUs), LAMAR selects between coarse-grained (CG) and fine-grained (FG) accesses based on workload patterns, reducing over-fetching and energy overhead. Key Benefits are reduced Over-fetching: Dynamically aligns memory fetch size to workload needs, optimizing bandwidth usage and saving energy. Energy Efficiency: Saves up to 38 percent memory energy for irregular workloads. Performance Gains: Efficiently balances irregular and regular workload phases, boosting overall performance.

5) *Graph Workloads and Irregular Memory Patterns*: Graph-based workloads like Breadth-First Search (BFS) suffer from irregular memory access patterns, limiting performance in traditional memory hierarchies. The Bi-modal Granularity Predictor (BGP) in LAMAR dynamically switches between CG and FG fetches, optimizing memory utilization. [9]

B. SIMT Efficiency and Control Flow Divergence

In Single Instruction, Multiple Threads (SIMT) architectures, control flow divergence poses a significant challenge to GPU performance. Divergence occurs when threads within a warp take different execution paths, leading to underutilized resources and increased computational stalls. Addressing these inefficiencies is crucial for improving throughput and scalability in diverse workloads. Here we have explored the following key techniques-

1) *Thread Block Compaction (TBC)*: Dynamically groups threads following the same execution path, reducing inactive threads within a warp.[9] Demonstrates up to a 22% speedup by improving warp utilization and minimizing idle cycles.

2) *Dynamic Warp Formation (DWF)*: Reassigns threads dynamically to new warps during execution to mitigate divergence. Particularly effective for nested control flows, though it introduces minor scheduling overhead.[9]

3) *Hybrid Techniques*: Combines thread compaction with advanced scheduling policies, such as age-based scheduling and round-robin prioritization, to optimize warp efficiency. Age-based scheduling enhances fairness and throughput, showing 15% improvement over traditional round-robin policies.

As shown in Figure 9, Thread Block Compaction dynamically tracks and reorganizes active threads within a warp, using the block-wide reconvergence stack and activemask. This mechanism enables higher warp utilization by minimizing idle cycles in divergent workloads.

bandwidth, reducing training times for large AI models like GPT-4 by up to 40%. [4] Energy Savings: Micron’s HBM3E achieves a 30% reduction in power consumption compared to competitors, underscoring its role in energy-efficient AI applications.[4]

2) **Thermal Management:** Innovations in packaging and bonding technologies, such as Molded Underfill (MR-MUF) and hybrid bonding, have significantly improved thermal resistance. These advancements enable safe scaling of HBM stacks, such as HBM3E, while maintaining performance in dense configurations.[12]

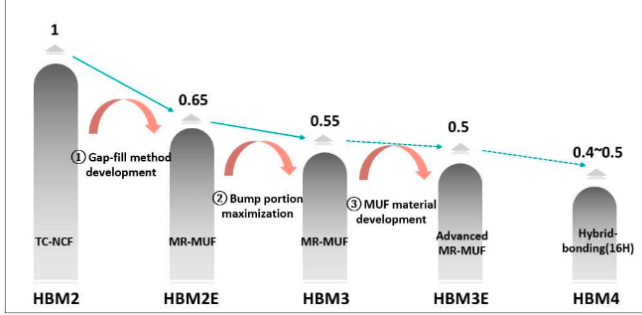


Fig. 8. Thermal resistance according to MR-MUF process improvement[12]

MR-MUF Technology: Enhances heat dissipation by optimizing thermal flow, ensuring stability in high-power regions.[12] **Hybrid Bonding:** Reduces thermal resistance and supports higher density HBM stacks, enabling efficient cooling for data-intensive workloads. Figure 7 shows thermal resistance improvements across HBM generations (HBM2 to HBM4) achieved through MR-MUF and hybrid bonding technologies.

3) **Chiplet and HBM Synergy:** **Localized Processing:** By processing data closer to memory, the combination of chiplets and HBM minimizes latency and energy usage.[7] **Scalability:** Modular chiplets allow customized configurations for diverse workloads, such as AI inference and graph analytics. **Performance Gains:** Studies demonstrate a 40% reduction in energy consumption when leveraging optimized chiplet configurations and locality-aware hierarchies. [7] These advancements highlight the critical role of HBM and chiplet synergy in improving energy efficiency and scalability, paving the way for next-generation GPUs optimized for AI and HPC applications.

IV. FUTURE TRENDS AND RESEARCH OPPORTUNITIES

The evolution of GPU architectures hinges on tackling challenges in scalability, energy efficiency, and workload adaptability. Future advancements will focus on hybrid memory systems, efficient cooling solutions, and smarter workload management.

Hybrid Memory Systems (HMS) offer a path forward by combining High Bandwidth Memory (HBM) with non-volatile technologies like 3D XPoint or NVRAM. This integration balances HBM’s speed with the scalability of SCM, dynamically allocating high-throughput tasks to HBM while using SCM for

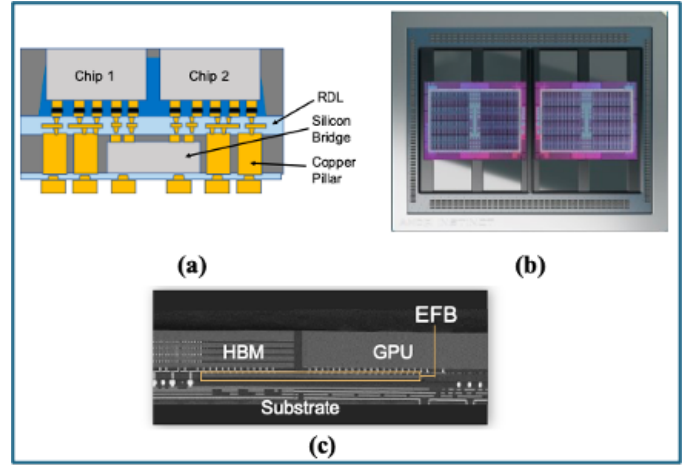


Fig. 9. (a) Cross-section schematic of two chips connected via a silicon bridge, (b) using silicon bridges to connect eight memory modules to two processor dies, and (c) a cross-sectional view of one of the bridge assemblies.[7]

less critical data. Addressing challenges such as data placement and memory coherence will require machine learning-driven algorithms to optimize performance and latency across hybrid layers.

Energy efficiency is another priority, particularly with the thermal bottlenecks posed by stacked HBM architectures. Advanced cooling methods, including micro-channel cooling, phase-change materials, and graphene-coated interposers, are enabling safe scaling of memory stacks like HBM3E [4]. AI-driven thermal management systems further enhance this by dynamically identifying and mitigating hotspots, ensuring stable performance in energy-constrained environments.

Efforts to improve SIMT efficiency will focus on reducing control flow divergence through techniques like Thread Block Compaction (TBC) and warp scheduling. Embedding these methods in hardware will cut overhead while improving throughput. AI-based schedulers that predict and adjust thread behavior in real time could further optimize resource allocation for complex workloads.

Finally, the adaptability of GPUs to emerging applications such as edge computing and AI inference will define their future role. Chiplet-based architectures combined with HBM enable low-latency, energy-efficient designs tailored to specific workloads, making them well-suited for real-time analytics and autonomous systems.

In summary, advancements in hybrid memory, thermal management, and intelligent scheduling will shape the next generation of GPUs. These innovations will address current limitations while unlocking new opportunities in AI, HPC, and edge computing, ensuring GPUs remain a cornerstone of modern computing.

V. CONCLUSION

This paper synthesizes advancements in GPU microarchitecture, emphasizing the transformative role of High Bandwidth Memory (HBM), data locality optimization, and energy-efficient designs. By addressing long-standing challenges in

memory bandwidth, computational efficiency, and scalability, the integration of HBM with emerging architectural paradigms—such as chiplet-based designs and hybrid memory systems—offers a pathway to sustained innovation. These technologies ensure that GPUs can meet the escalating demands of AI, HPC, and real-time analytics, all while minimizing energy costs and maximizing resource utilization.

In addition to highlighting the benefits of data locality-aware hierarchies and SIMT optimization techniques, this study provides a foundation for tackling emerging challenges like thermal constraints and workload variability. By bridging current advancements with practical applications, this paper underscores GPUs' critical role in shaping the future of computing across domains such as AI, autonomous systems, and edge computing. These innovations, coupled with a forward-looking perspective on energy and architectural efficiency, position GPUs to remain indispensable in the evolving landscape of data-intensive applications.

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